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Our customers span over 70 countries and cover diverse markets and industries.



Kinco® Automation

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Kinco

Motion Control
Servo System

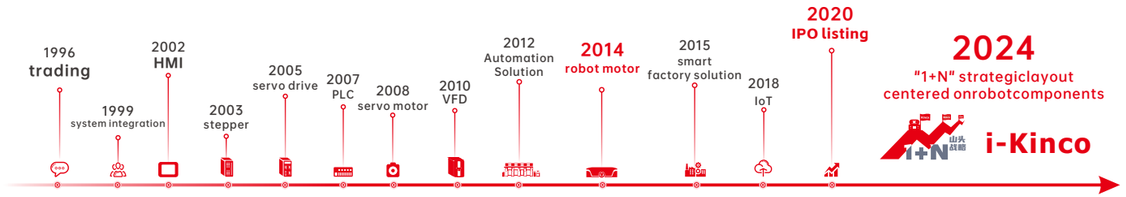
Servo System Catalog

- FD6P Servo Driver
- SMK Servo Motor



About us





Kinco was founded in 1996, and successfully listed on the Shanghai Stock Exchange in 2020 (abbreviated name: Kinco share, stock code 688160), which is a high-tech, specialized and sophisticated enterprise that attaches great importance to independent research and development and innovation, mainly engaged in the research and development, production, sales and related technical services of industrial automation and robot core components and digital factory hardware and software.Leading Chinese supplier of motion control and human-machine interaction components for robotics and intelligent devices.

After years of continuous research and development and innovation, Kinco has established a complete product line with independent intellectual property rights, covering a series of products from machine iot to human-machine interaction, control, drive and execution, which are widely used in robots, medical equipment, logistics equipment, packaging equipment, food equipment, clothing equipment, environmental protection equipment, etc. New energy equipment, rail transit equipment and other automation equipment industry.

Based on the comprehensive industrial automation and digital technology platform, the company has in-depth application scenarios in the robot industry, providing display, control, drive and other multi-dimensional solutions for industrial mobile robots, collaborative robots, industrial robots, pan-service robots, and bionic robots. Through the insight of the industry pain points, deep links with robot customers, combined with the advantages of product research and development, the company continues to innovate, and launches industry-leading low-voltage servo products for mobile robots, integrated servo wheel, frameless torque motor for collaborative robots, robot human-machine interfaces, robot controllers and other products. The company has formed a relatively complete robot core parts capability, and after nearly 10 years of hard work in the robot industry, it has become a leading enterprise in the field of mobile robot low-voltage servo, and has a high brand influence in the industry.

Kinco has four research and development centers in Shanghai, Shenzhen, Changzhou and Chengdu, and two manufacturing bases in Shenzhen and Changzhou, a total of 10+ domestic marketing centers, 100+ domestic service providers, 40+ global partners, and products are exported to 70+ countries overseas. In terms of after-sales service, Kinco has established after-sales service centers in Shanghai, Shenzhen and Changzhou.

Four R&D centers and two manufacturing bases



Shanghai | Shenzhen | Changzhou | Chengdu | Shenzhen Manufacturing Center HMI, PLC, and drive systems | Changzhou Manufacturing Base Motor

Kinco High voltage servo system

The Kinco FD6P is a high-performance AC servo drive designed for general-purpose applications. With a power range of 50W~3kW, it is fully compatible with SMK series servo motors. It supports multiple mainstream communication protocols, including EtherCAT and Modbus, ensuring excellent compatibility and flexible integration with various host control systems.

Featuring easy commissioning, high safety, and stable performance, the FD6P helps shorten equipment development cycles and reduce system integration complexity. It is widely used in automation applications such as woodworking, packaging, logistics, and food processing, providing users with a reliable and cost-efficient motion control solution.



EtherCAT Modbus

Power planning covers 50W-3kW;

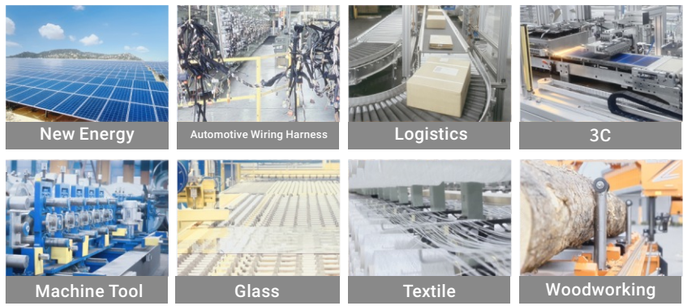
High responsiveness, speed loop sampling frequency up to 16KHz;

Miniaturized body;

Supports parameter self-tuning;

Supports S-curve control, end vibration suppression, high-speed probe, and other functions.

Industry application



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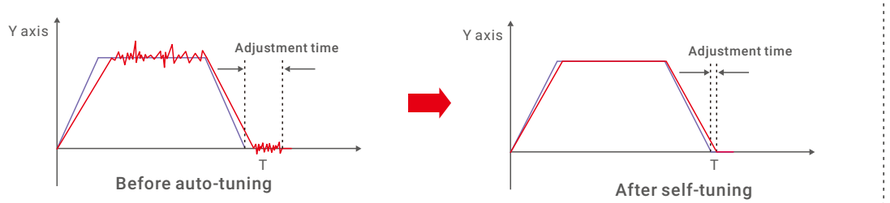
Seven Advantages

1 Vibration suppression and high-performance self-tuning

The built-in command filter algorithm and notch filter effectively suppress mechanical resonance and operating jitter, improving system operation stability and processing accuracy.

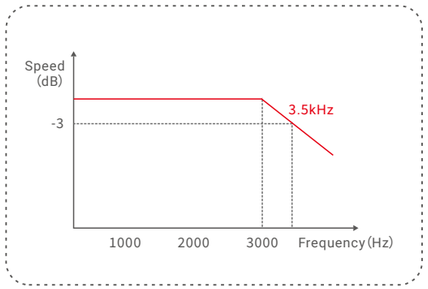


High-performance one-click auto-tuning simplifies gain debugging. The servo system automatically completes parameter optimization after running, significantly improving tuning efficiency.



2 High-performance control core

The speed loop bandwidth is increased to 3.5kHz, combined with the inertia recognition performance optimization , to achieve high-response and high-precision dynamic control to meet the stringent requirements in complex working condition.



3 Intelligent debugging and convenient operation and maintenance

Supports Bluetooth debugging and cooperates with the new KS Pro software to achieve wireless connection, parameter configuration and status monitoring;

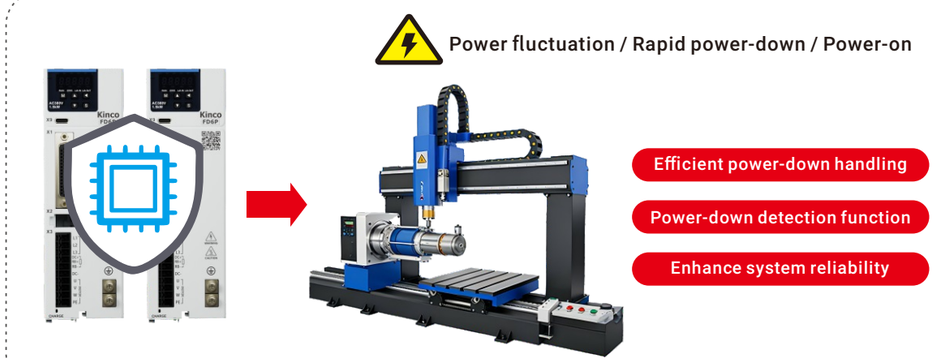
Equipped with 5-digit buttons, it provides a new panel debugging experience and simplifies the on-site operation process.



Seven Advantages

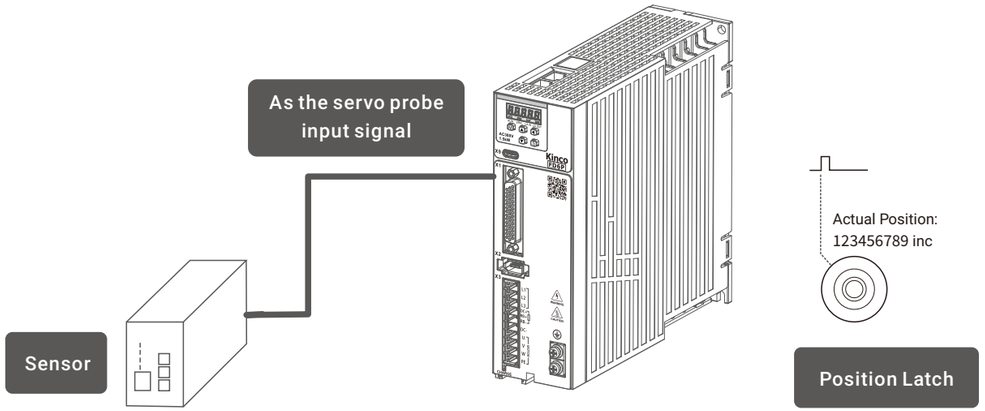
4 Efficient power-down handling

Equipped with power-down detection function, it can meet the system stability requirements in rapid power-down and power-on scenarios, and improve the reliability of the device in power supply fluctuation environment.



5 Efficient signal processing

Supports high-speed probe function to achieve microsecond-level signal capture and response, suitable for time-sensitive application scenarios such as high-precision positioning and high-speed triggering.

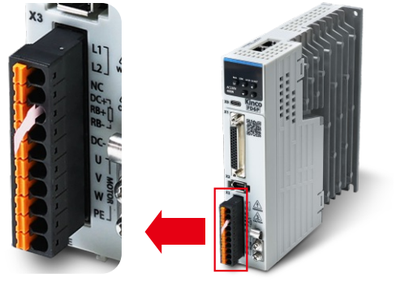


Seven Advantages

6 Strengthen hardware reliability

Models with power of 750W and above are equipped with a braking resistor as standard to improve braking capacity and system safety; the power connector adopts a high-reliability design to enhance anti-vibration and anti-interference capabilities, making it suitable for harsh industrial environments.

Strengthen hardware reliability, anti-vibration, and anti-interference



Power terminal

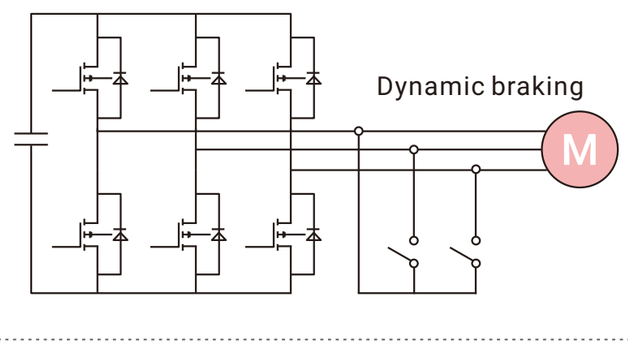
Improved braking capability + system safety



Braking resistor

7 Dynamic braking

Sudden failure can still shut down smoothly, effectively preventing equipment collision damage. It is suitable for high-speed running equipment such as engraving machines and cutting machines. When an alarm is triggered or there is an unexpected power outage, the motor can brake and stop quickly and smoothly.



Note: The STO model is currently under development. Stay tuned for updates!

SMK features

Characteristics of SMK servo motor

New electromagnetic design

Brand-new electromagnetic solution, with small slot torque and low torque pulsation,which is conducive to reducing the vibration during the operation of the motor and making the torque output more smooth;

New structure and short fuselage

The redesign of the fuselage structure shortens the length of the fuselage, which can save more installation space and reduce the size of the equipment for customers' equipment;

Insulation class F

The motor in the industry is at the highest insulation level, which can maintain high reliability and stability in high temperature extreme environment;

Energy efficiency: Level 1



Naming rule

Drive: FD6P-E A-L 2R9 -XXX

① ② ③ ④ ⑤ ⑥



①-Series Name	FD6P:FD6P series	⑤-Drive current	2R9:2.9A 4R0:4A 5R4:5.4A 5R5:5.5A 8R6: 8.6A 013:13A
②-Control mode	L:RS485，Pulse E:EtherCAT		
③- Hardware series	A:Standard hardware	⑥- Special code	Empty: Standard version XXX: Custom code,Can be freely combined with numbers and letters (0-10 digits)
④- Supply voltage	L:AC220V H: AC380V		

Naming rule

Motor: SMK 60 S - 0040 - 30 Y A K - 5 L S R
① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩ ⑪ ⑫



①-Series Name	SMK: SMK series
②-Flange	40:40×40（mm） 60:60x60(mm) 80:80x80(mm) 130:130x130(mm) 180:180x180(mm)
③-Inertia type	S: Small inertia G: Large inertia
④-Rated power	0005:10x5(W) 0010:10x10(W) 0040:10x40(W) ..... 0300:10x300(W)
⑤-Rated speed	15:15x100(rpm) 30:30x100(rpm)
⑥-Encoder type	Y:23 bit multi-turn communication optical absolute value encoder
⑦-Brake	A:Without brake B:With brake
⑧-Output shaft style	K:With key
⑨-Number of polar pairs	5:5-pole pair
⑩- Supply voltage	L:AC220V H:AC380V
⑪- Motor version number	S:S version
⑫- Motor Outlet Type	R:Standard socket

Encoder cable: E - D - QY - K A0 - LL - F
① ② ③ ④ ⑤ ⑥ ⑦

①-Cable function type	E:Motor encoder cable
②-Driver end encoder connector type	D:1394 6-core connector
③-Encoder type	QY:Multiturn communication encoder
④ -Motor platform	K:SMK
⑤-Motor end connector	A0：Common body motor front outgoing. A1：Common body motor rear outgoing. R：Military plug rear outgoing. F：Military plug front outgoing.
⑥-Cable length(m)	3:3m 5:5m 10:10m
⑦-Cable property	F:High flexibility cable

Naming rule

Brake extension cable: BRA - EXT - LL
① ② ③

①-Cable function type	BRA:Motor brake extension cable
②-Drive end connector	EXT:Extension connector inside the cabinet
③-Cable length(m)	0.5:0.5m 3:3m 5:5m

Power cable: M - A - 6A - K A0 - LL - BF
① ② ③ ④ ⑤ ⑥ ⑦

①-Cable function type	M:Motor power cable
②-Drive end connector	A:Needle type cold press terminal Y:Fork shaped cold press head
③-Current	6A:Rated current 6A 12A:Rated current 12A 20A:Rated current 20A
④ -Motor platform	K:SMK
⑤-Motor end connector	A0：Common body motor front outgoing. A1：Common body motor rear outgoing. R0：130 Military plug rear outgoing. F0：130 Military plug front outgoing. R1：180 Military plug rear outgoing. F1：180 Military plug front outgoing.
⑥-Cable length(m)	3:3m 5:5m 10:10m
⑦-Cable property	F:High flexibility cable FS:High flexibility shielded cable BF:High flexible cable with brake BFS:High flexibility shielded cable with brake

FD6P driver and SMK motor configuration table

Servo motor			Servo drive		Brake cable		Power cable(Note 2)			Encoder cable			
Inertia type	Rated parameter	Motor model	Pulse+Modbus	EtherCAT	Extension cable inside the cabinet (Note 1)		Axial out line	Tail out line	Plug accessory kit	Axial out line	Tail out line	Encoder cable (Note 3)	Plug accessory kit
Small inertia	220V/50W/0.88A/ 3000rpm/0.16Nm	SMK40S-0005-30YAK-5LSR	FD6P-LA-L2R9	FD6P-EA-L2R9	BRA-EXT-LL		M-A-6A-KA0-LL-F/(-FS) M-A-6A-KA0-LL-BF/(-BFS)	M-A-6A-KA1-LL-F/(-FS) M-A-6A-KA1-LL-BF/(-BFS)	M-KA0 M-KA1	E-D-QY-KA0-LL-F	E-D-QY-KA1-LL-F	BAT-FD5	E-D E-KA
		SMK40S-0005-30YBK-5LSR											
	220V/100W/1.2A/ 3000rpm/0.32Nm	SMK40S-0010-30YAK-5LSR											
		SMK40S-0010-30YBK-5LSR											
	220V/200W/1.55A/ 3000rpm/0.64Nm	SMK60S-0020-30YAK-5LSR											
		SMK60S-0020-30YBK-5LSR											
	220V/400W/2.93A/ 3000rpm/1.27Nm	SMK60S-0040-30YAK-5LSR											
		SMK60S-0040-30YBK-5LSR											
	220V/750W/3.9A/ 3000rpm/2.39Nm	SMK80S-0075-30YAK-5LSR	FD6P-LA-L4R0	FD6P-EA-L4R0									
		SMK80S-0075-30YBK-5LSR											
	220V/1000W/5.3A/ 3000rpm/3.18Nm	SMK80S-0100-30YAK-5LSR	FD6P-LA-L5R5	FD6P-EA-L5R5									
		SMK80S-0100-30YBK-5LSR											
Large inertia	220V/850W/5.4A/ 1500rpm/5.39Nm	SMK130G-0085-15YAK-5LSR	FD6P-LA-L5R5 FD6P-LA-H5R4	FD6P-EA-L5R5 FD6P-EA-H5R4									
		SMK130G-0085-15YBK-5LSR											
	220V/1300W/7.5A/ 1500rpm/8.27Nm	SMK130G-0130-15YAK-5LSR					M-A-12A-KR0-LL-F/(-FS) M-A-12A-KR0-LL-BF/(-BFS)	M-A-12A-KF0-LL-F/(-FS) M-A-12A-KF0-LL-BF/(-BFS)	M-KR0	E-D-QY-KR-LL-F	E-D-QY-KF-LL-F	BAT-FD5	E-D E-KR
		SMK130G-0130-15YBK-5LSR	FD6P-LA-H8R6	FD6P-EA-H8R6									
	220V/1800W/10A 1500rpm/11.45Nm	SMK130G-0180-15YAK-5LSR	FD6P-LA-H013	FD6P-EA-H013	BRA-EXT-LL								
		SMK130G-0180-15YBK-5LSR											
	220V/2400W/12.9A/ 1500rpm/15.2Nm	SMK130G-0240-15YAK-5LSR											
		SMK130G-0240-15YBK-5LSR											
	380V/3000W/10.6A/ 1500rpm/19.1Nm	SMK180G-0300-15YAK-5HSR					M-A-12A-KR1-LL-F/(-FS) M-A-12A-KR1-LL-BF/(-BFS)	M-A-12A-KF1-LL-F/(-FS) M-A-12A-KF1-LL-BF/(-BFS)	M-KR1				
		SMK180G-0300-15YBK-5HSR											

(Note1): The extension cable inside the cabinet is used to match the power cable with a brake

(Note2): The parentheses behind the power cable indicate the suffix of another cable

(Note3): Y encoders are used when multi-turn function is required

FD6P Servo drive technical parameters table

Model Parameter		FD6P Servo		
	Drive model	FD6P-▲A-L2R9	FD6P-▲A-L4R0	FD6P-▲A-L5R5
Power supply	Power	1PH，200-240VAC±10% ，50/60Hz±3HZ		
	Logic power	Null	Null	Null
Current	Max continuous output current（Arms）	2.9	4	5.5
	Peak current（Arms）	10.61	12.73	16.97
	Rated output power Pn(W)	400	750	1000
	Feedback signal	Communication type encoder	Communication type encoder	Communication type encoder
	Energy consumption brake	No built-in brake resistance	Built-in brake resistance 100Ω，limited power 60W	
		Note: the actual power more than limit power, need external brake resistor (according to operation, mainly used in the rapid start /stop occasion)		
	Energy consumption braking voltage absorption point	DC380V±5V	DC380V±5V	DC380V±5V
	Overvoltage alarm voltage	DC400V±5V	DC400V±5V	DC400V±5V
	Undervoltage alarm voltage	DC200V±5V	DC200V±5V	DC200V±5V
	Cooling method	Natural cooling	Natural cooling	Forced air cooling
	Weight（Kg）	0.824	1.166	1.166
	Encoder signal output function	Output 5V motor A, B, Z signals, frequency division output range 0 ~ 65536; used for multi-axis synchronization, maximum output frequency 5MHz Note :EA model does not have encoder output signal functionality.		
	Protection function	Over-voltage protection, under-voltage protection, motor overheating (I^2T) protection, short-circuit protection, drive overheating protection, etc.		
	Type-C	For debugging purposes only		
	RS485	Maximum support 115200 bps baud rate, can use Modbus RTU protocol to communicate with the controller		
	Ethercat	Support CoE (CiA402 protocol) and CSP/CSV/PP/PV/PT/HM mode, communication rate 100M		
Location control mode	Command control mode	External pulse input control; Control of 8-segment position using DIN signal; Communication setting internal object parameter control		
	Command smoothing mode	Low-pass filtering (set by internal parameters), S-curve smoothing filtering (set by internal parameters in 1 mode)		
	Pulse command mode	Pulse+direction, CCW+CW, A-phase+B-phase		
	Maximum input pulse frequency	Differential transmission mode: up to 4MHz, open collector transmission mode: 500KHz		
	Electronic gear ratios	Setting range Gear factor:-32768～32767，Gear divider:1～32767，1/50≤|Gear factor/Gear divider|≤50		
	Torque limit	Internal parameter setting		
	Feedforward gain	0~100.0%（Internal parameter setting）		
	Position loop sampling frequency	8KHz		
Speed control mode	Command control mode	8-segment speed control using DIN signals，Communication settings internal object parameter control		
	Command smoothing mode	Low-pass filtering（Internal parameter setting）		
	Speed limit	Internal parameter setting		
	Torque limit	Internal parameter setting		
	Speed loop sampling frequency	16KHz		
Torque control mode	Command control mode	Communication setting internal object parameter control		
	Command smoothing mode	Low-pass filtering（Internal parameter setting）		
	Speed limit	Internal parameter setting		
	Current loop sampling frequency	16KHz		
Digital input	Input specification	L model: 7 digital input, E model: 5 digital input Effective input voltage range: 12.5-30V via COMI terminal connection Note: DI2-DI3 are high-speed digital inputs (maximum input frequency: 10KHz)		
	Input function	The following functions can be defined according to your needs: drive enable, drive error reset, drive working mode control, speed loop proportional control, positive limit, negative limit, origin signal, command reverse, internal speed segment control, internal position segment control, emergency stop, pause, start to find the origin, command activation, wheel ratio switching, gain switching, position table function, clear pulse function, etc		
Digital output	Output specification	L model: 5 digital output, E model: 3 digital output Max voltage : DC30V ,Max output current: 30mA ,the DO2 brake output is controlled by a relay		
	Output function	The following functions can be defined according to your needs:drive ready, drive error, motor position arrives, motor zero speed, motor holding brake, motor speed arrives, index Z signal appears, speed reaches limit, torque reaches setting, motor lock Axis, motor limit, origin found, multi-segment position, etc		
Operation environment	Operation temperature	-10°C ~ +40°C 40°C ~ 55°C (When the ambient temperature exceeds 40°C, derate by 3% for every 1°C increase.)		
	Storage temperature	-30°C ~ +70°C (Does not freeze)		
	Humidity	Below 90%RH (no condensation)		
	Protection level	IP20 Note: Excluding terminals(IP00)		
	Installation site	Dust-free, dry, lockable (e.g. electrical cabinets)		
	Installation method	Vertical installation		
	Installation altitude	Less than 2000m,power derating 1.5%/100m between 1000m and 2000m		
	Atmosphere pressure	86kpa～106kpa		
	Vibration	5Hz~ 55Hz，0.35mm bidirectional displacement 55Hz~ 500Hz，2g constant acceleration		

Note：▲= L: RS485，pulse
E: EtherCAT

FD6P Servo drive technical parameters table

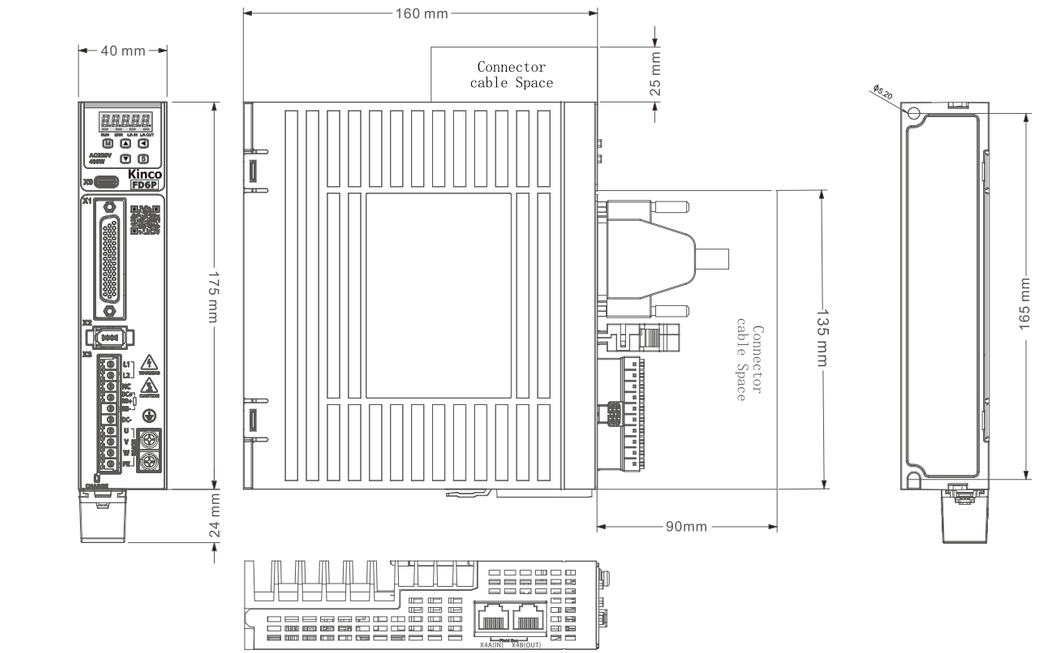
Model Parameter		FD6P Servo		
	Drive model	FD6P-▲A-H5R4	FD6P-▲A-H8R6	FD6P-▲A-H013
Power supply	Power	3PH,380V-415VAC±10%，50/60Hz±3H2		
	Logic power	Null	1PH，380V-415VAC±10%，50/60Hz±3Hz	
Current	Max continuous output current（Arms）	5.4	8.6	13
	Peak current（Arms）	16.97	22.98	32.53
	Rated output power Pn(W)	1500	2000	3000
	Feedback signal	Communication type encoder	Communication type encoder	Communication type encoder
	Energy consumption brake	Built-in brake resistance 100Ω，limited power 60W		
		Note: the actual power more than limit power, need external brake resistor (according to operation, mainly used in the rapid start /stop occasion)		
	Energy consumption braking voltage absorption point	DC680V±5V	DC680V±5V	DC680V±5V
	Overvoltage alarm voltage	DC710V±5V	DC710V±5V	DC710V±5V
	Undervoltage alarm voltage	DC400V±5V	DC400V±5V	DC400V±5V
	Cooling method	Forced air cooling	Forced air cooling	Forced air cooling
	Weight（Kg）	1.271	1.931	1.931
	Encoder signal output function	Output 5V motor A, B, Z signals, frequency division output range 0 ~ 65536; used for multi-axis synchronization, maximum output frequency 5MHz Note :EA model does not have encoder output signal functionality.		
	Protection function	Over-voltage protection, under-voltage protection, motor overheating (I^2T) protection, short-circuit protection, drive overheating protection, etc.		
	Type-C	For debugging purposes only		
	RS485	Maximum support 115200 bps baud rate, can use Modbus RTU protocol to communicate with the controller		
	Ethercat	Support CoE (CiA402 protocol) and CSP/CSV/PP/PV/PT/HM mode, communication rate 100M		
Location control mode	Command control mode	External pulse input control; Control of 8-segment position using DIN signal; Communication setting internal object parameter control		
	Command smoothing mode	Low-pass filtering (set by internal parameters), S-curve smoothing filtering (set by internal parameters in 1 mode)		
	Pulse command mode	Pulse+direction, CCW+CW, A-phase+B-phase		
	Maximum input pulse frequency	Differential transmission mode: up to 4MHz, open collector transmission mode: 500KHz		
	Electronic gear ratios	Setting range Gear factor:-32768～32767，Gear divider:1～32767，1/50≤|Gear factor/Gear divider|≤50		
	Torque limit	Internal parameter setting		
	Feedforward gain	0~100.0%（Internal parameter setting）		
	Position loop sampling frequency	8KHz		
Speed control mode	Command control mode	8-segment speed control using DIN signals，Communication settings internal object parameter control		
	Command smoothing mode	Low-pass filtering（Internal parameter setting）		
	Speed limit	Internal parameter setting		
	Torque limit	Internal parameter setting		
	Speed loop sampling frequency	16KHz		
Torque control mode	Command control mode	Communication setting internal object parameter control		
	Command smoothing mode	Low-pass filtering（Internal parameter setting）		
	Speed limit	Internal parameter setting		
	Current loop sampling frequency	16KHz		
Digital input	Input specification	L model: 7 digital input, E model: 5 digital input Effective input voltage range: 12.5-30V via COMI terminal connection Note: DI2-DI3 are high-speed digital inputs (maximum input frequency: 10KHz)		
	Input function	The following functions can be defined according to your needs: drive enable, drive error reset, drive working mode control, speed loop proportional control, positive limit, negative limit, origin signal, command reverse, internal speed segment control, internal position segment control, emergency stop, pause, start to find the origin, command activation, wheel ratio switching, gain switching, position table function, clear pulse function, etc		
Digital output	Output specification	L model: 5 digital output, E model: 3 digital output Max voltage : DC30V ,Max output current: 30mA ,the DO2 brake output is controlled by a relay		
	Output function	The following functions can be defined according to your needs:drive ready, drive error, motor position arrives, motor zero speed, motor holding brake, motor speed arrives, index Z signal appears, speed reaches limit, torque reaches setting, motor lock Axis, motor limit, origin found, multi-segment position, etc		
Operation environment	Operation temperature	-10°C ~ +40°C 40°C ~ 55°C (When the ambient temperature exceeds 40°C, derate by 3% for every 1°C increase.)		
	Storage temperature	-30°C ~ +70°C (Does not freeze)		
	Humidity	Below 90%RH (no condensation)		
	Protection level	IP20 Note: Excluding terminals(IP00)		
	Installation site	Dust-free, dry, lockable (e.g. electrical cabinets)		
	Installation method	Vertical installation		
	Installation altitude	Less than 2000m,power derating 1.5%/100m between 1000m and 2000m		
	Atmosphere pressure	86kpa～106kpa		
	Vibration	5Hz~ 55Hz，0.35mm bidirectional displacement 55Hz~ 500Hz，2g constant acceleration		

Note：▲= L: RS485，pulse
E: EtherCAT

Servo drive mechanical dimension

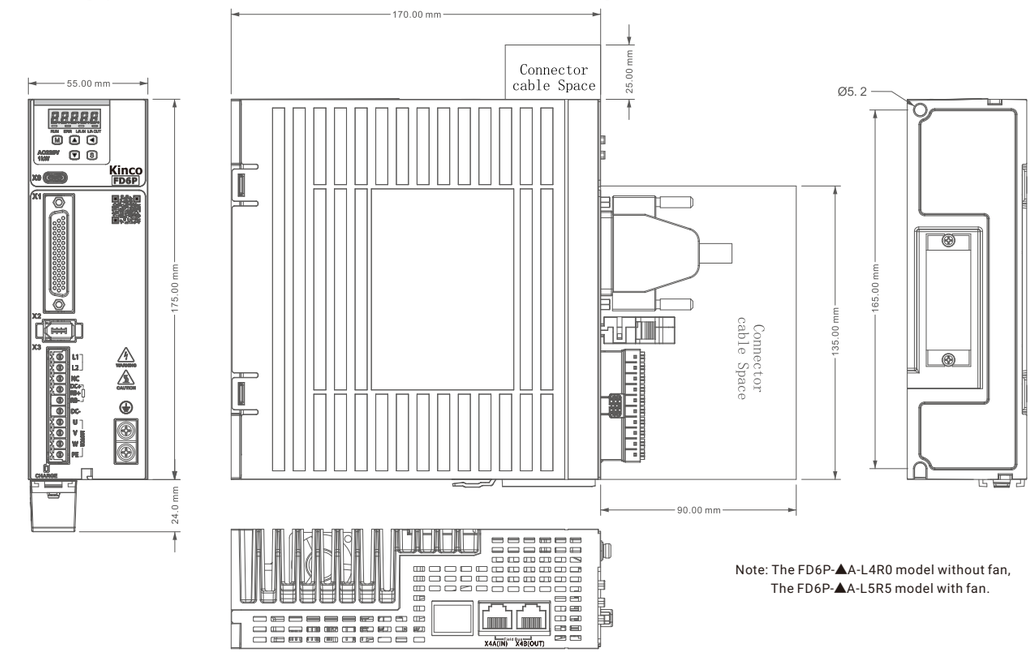
FD6P-▲A-L2R9 (Unit:mm)

Note: Wiring space needs to be reserved around the driver, it is recommended to be greater than 70mm.



FD6P-▲A-L4R0/L5R5 (Unit:mm)

Note: Wiring space needs to be reserved around the driver, it is recommended to be greater than 70mm.

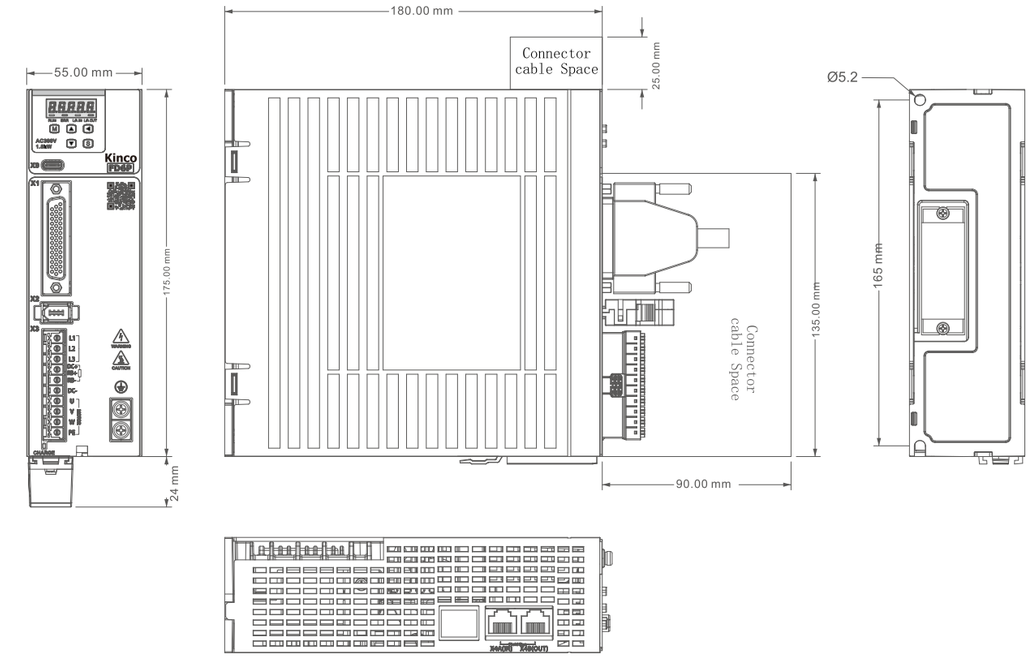


Note: The FD6P-▲A-L4R0 model without fan,
The FD6P-▲A-L5R5 model with fan.

Servo drive mechanical dimension

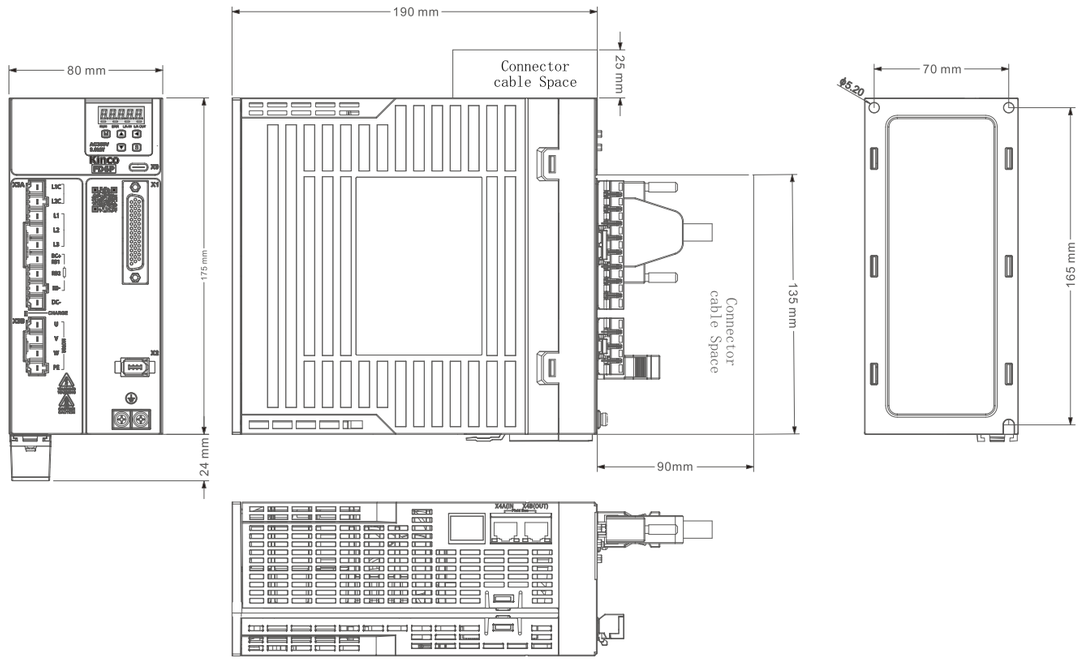
FD6P-▲A-H5R4 (Unit:mm)

Note: Wiring space needs to be reserved around the driver, it is recommended to be greater than 70mm.



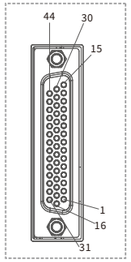
FD6P-▲A-H8R6/H013 (Unit:mm)

Note: Wiring space needs to be reserved around the driver, it is recommended to be greater than 70mm.

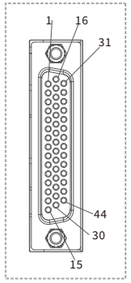


Driver terminal description and wiring

Interface definition



FD6P-▲A-L2R9/L4R0/
L5R5/H5R4



FD6P-▲A-/H8R6/H013

X1
I/O interface

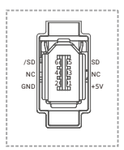
Pin	Signal	Signal name	Specification
7	DO1+	Differential output	Open collector output MAX voltage DC30V MAX current 30mA
6	DO1-		
5	DO2+		
4	DO2-		
3	DO3+		
2	DO3-		
1	DO4+		
26	DO4-		
28	DO5+		
27	DO5-		
11	COMI	Signal input port common terminal	Digital input port common terminal,Power supply range:12.5~30VDC
9	DI1	Digital signal input	High level:12.5V~30V Low level:0~5V Max input frequency:10KHz
10	DI2 (Fast)		
34	DI3 (Fast)		
8	DI4		
33	DI5		
32	DI6		
12	DI7		
35	PULL HI	24V pulse input interface	The pulse input 24V interface can be used as the negative terminal with PLS +and SIGN+, or as the positive terminal with PLS- and SIGN-.
41	PLS+	Low-frequency pulse input interface	Low-frequency pulse signal input terminal, maximum 500kHz, voltage range DC3.3~5V
43	PLS-		
37	SIGN+		
39	SIGN-		
38	HPLS+	High-frequency pulse input interface	High-frequency pulse signal output terminal, maximum 4MHz, voltage range DC 3.3~5V
36	HPLS-		
42	HSIGN+		
40	HSIGN-		
21	ENCO_A	Encoder signal output	Output 5V A, B, Z differential signal, the frequency division output range is 0~65536; for multi-axis synchronization, the maximum output frequency is 5MHz
22	ENCO_A/		
23	ENCO_B/		
25	ENCO_B		
13	ENCO_Z		
24	ENCO_Z/		
44	OCZ	Motor zero output	Refer to the OC output at the GND terminal,The maximum support is DC30V and the maximum current is 30mA
29	GND	Internal reference GND of the drive	Internal reference GND of the drive
17	VDD	24V+	Internal 24V power supply output, voltage range +/-20%，Max current
14	VEE	24V-	200mA

Note 1：L model: 7 digital input DI1~DI7，5 digital output D01~D05；
E model: 5 digital input DI1~DI5(PIN32、12 are empty)，3 digital output D01~D03(Pin 1, 26, 27, 28 are empty)；
Note 2：The EA model does not have pulse or encoder output signal functionality.



TYPE-C
USB interface

Pin	Signal	Signal name
A4 / B4	VBUS	USB power
A9 / B9	VBUS	
A1 / B1	GND	USB power ground
A12 / B12	GND	
A6 / B6	DP+	Differential data signal
A7 / B7	DP-	
A5	CC1	Detection pin, internally pulled down
B5	CC2	

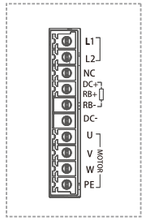


X2
Encoder
interface

Pin	Signal	Signal name	Specification
1	+5V	5V power supply output positive	Encoder signal input port
2	GND	5V power supply output positive	
3	CLOCK+	Clock signal positive	
4	CLOCK-	Clock signal negative	
5	SD	Data signal	
6	/SD		

Driver terminal description and wiring

Interface definition

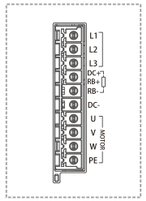


FD6P-▲A-L2R9/4R0/5R5

X3
Power
interface

Signal		Specification
L1		Drive the main power input
L2		Single-phase line voltage: 200～240VAC ±10%, 50～60Hz ±3Hz
NC		Empty pin, not allowed to connect
DC+/RB+	DC+	DC bus positive
	RB+	Brake resistor positive
RB-		Brake resistor negative
DC-		DC bus negaitive
U/V/W		Servo motor UVW phase cable connection
PE		Ground terminal

Note:The L2R9 model does not have a built-in braking resistor.
The L4R0/L5R5 model are equipped with a small aluminum shell resistor from the factory, which is connected between RB+ and RB-.
When the braking power exceeds this resistor, it needs to be removed and a larger power resistor must be connected externally.

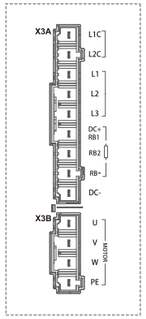


FD6P-▲A-H5R4

X3
Power
interface

Signal		Specification
L1/L2/L3		Power supply input Three-phase line voltage:380～415VAC ±10%, 50～60Hz ±3Hz
DC+ /RB+	DC+	DC bus positive
	RB+	Brake resistor positive
RB-		Brake resistor negative
DC-		DC bus negaitive
U/V/W		Servo motor UVW phase cable connection
PE		Ground terminal

Note:The H5R4 model are equipped with a small aluminum shell resistor from the factory,
which is connected between RB+ and RB-. When the braking power exceeds this resistor,
it needs to be removed and a larger power resistor must be connected externally.

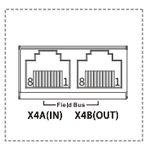


FD6P-▲A-H8R6/H013

X3
Power
interface

Port	Signal		Specification
X3A	L1C/L2C		Logic power input Single-phase line voltage: 380～415VAC ±10%, 50～60Hz ±3Hz
	L1/L2/L3		Power supply input Three-phase line voltage:380～415VAC ±10%,
	DC+ /RB1	DC+	50～60Hz ±3Hz
		RB1	DC bus positive
	RB2		When using internal brake resistor, RB1 and RB2 are short circuited
	RB-		When using an external brake resistor, disconnect the short cable between RB1 and RB2 , and connect the external brake resistor to RB1 and RB-
	DC-		DC bus negaitive
X3B	U/V/W		Servo motor UVW phase cable connection
	PE		Ground terminal

Note:The H8R6/H013 model model are equipped with a small aluminum shell resistor from the factory, which is connected between RB1and RB2



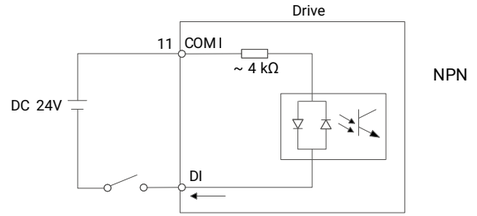
X4
Communication
interface

Pin	RS485	EtherCAT
1	-	TD+
2	-	TD-
3	-	RD+
4	485-	-
5	485+	-
6	-	RD-
7	-	-
8	GND_RS485	-

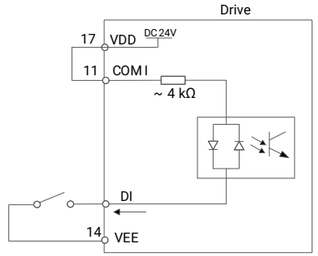
Digital input and output wiring

Digital signal input wiring

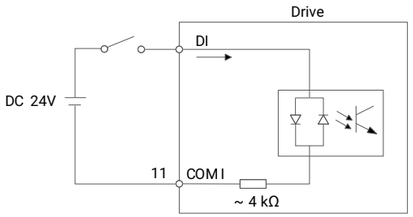
Maximum input frequency for digital input port: 10 kHz;Effective input level: 12.5–30 V via COMI terminal connection
Internal 24 V output can be used as pull-up power supply



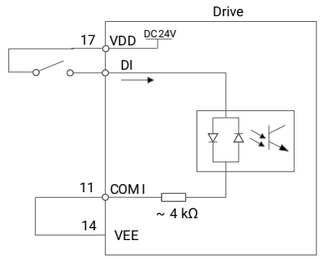
NPN



NPN



PNP



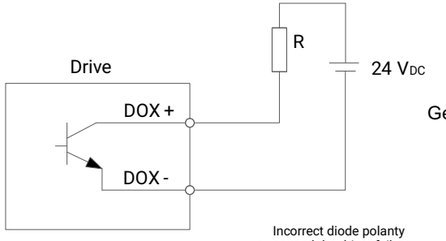
PNP

Using external power supply

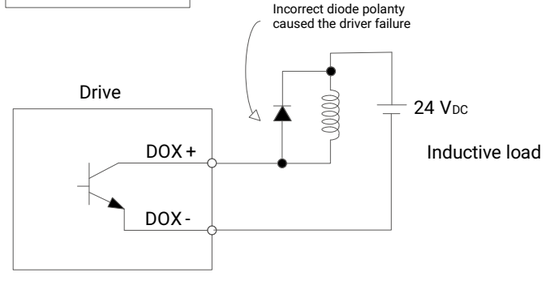
Use internal power supply

Digital signal output wiring

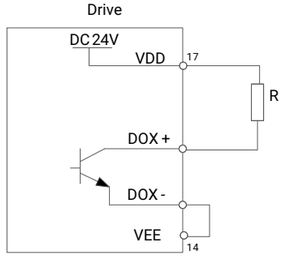
Digital output port maximum voltage DC30V, maximum current 30mA
The internal 24V output of the driver can be used as the pull-up power supply.



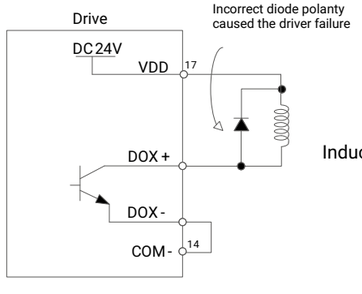
General load



Inductive load



General load



Inductive load

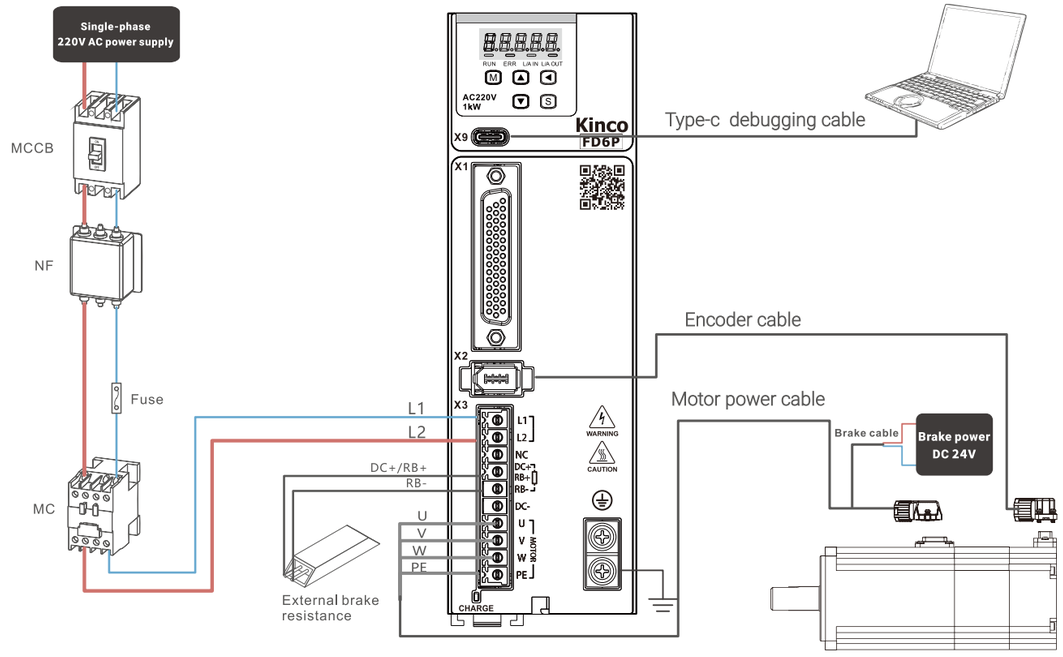
Using external power supply

Use internal power supply

Drive external wiring diagram

External wiring diagram

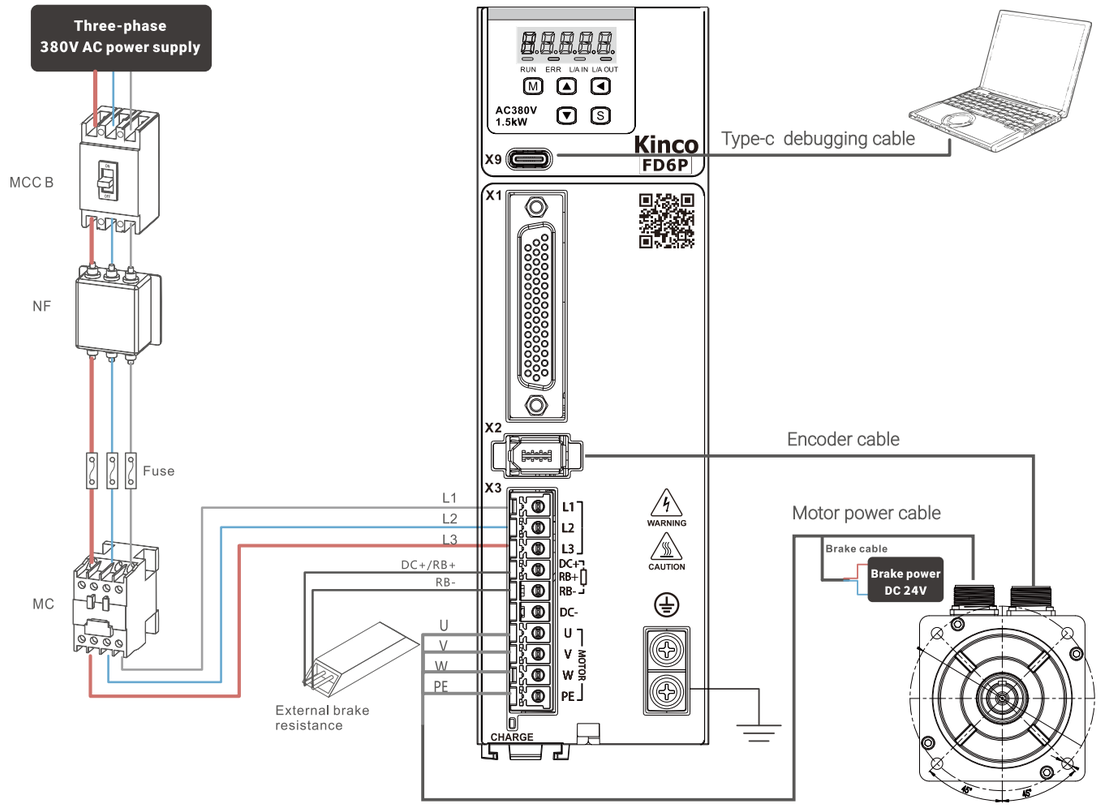
FD6P-▲A-L2R9/4R0/5R5



Drive external wiring diagram

External wiring diagram

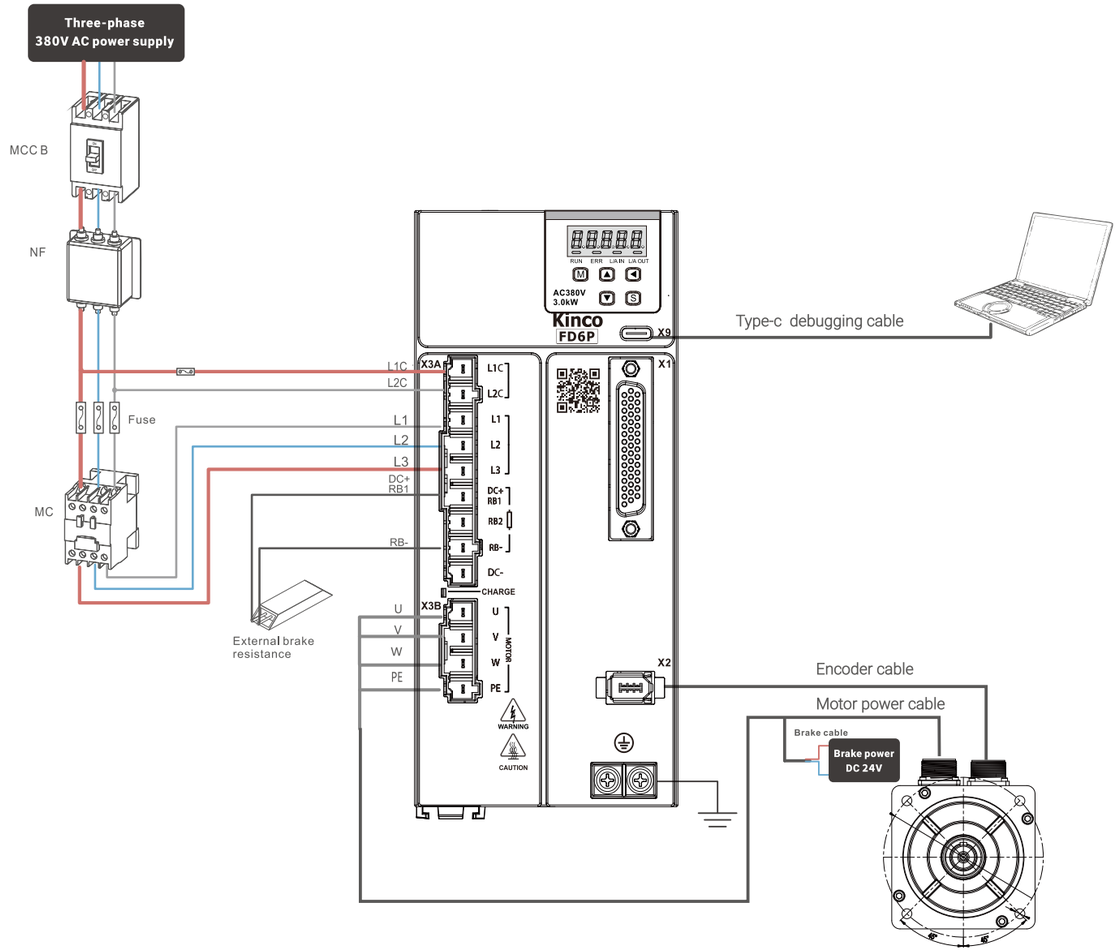
FD6P-▲A-H5R4



Drive external wiring diagram

External wiring diagram

FD6P-▲A-H8R6/H013



SMK servo motor technical parameter

Model parameter		Small inertia，40 flange		Small inertia，60 flange		Small inertia，80 flange	
Servo motor model		SMK40S-0005 -30Y■K-5LSR	SMK40S-0010 -30Y■K-5LSR	SMK60S-0020 -30Y■K-5LSR	SMK60S-0040 -30Y■K-5LSR	SMK80S-0075 -30Y■K-5LSR	SMK80S-0100 -30Y■K-5LSR
Adapted drives		FD6P-▲A-L2R9	FD6P-▲A-L2R9	FD6P-▲A-L2R9	FD6P-▲A-L2R9	FD6P-▲A-L4R0	FD6P-▲A-L5R5
Rated voltage（VAC）		220	220	220	220	220	220
Rated power Pn(W)		50	100	200	400	750	1000
Rated torque Tn(N.m)		0.16	0.32	0.64	1.27	2.39	3.18
Rated speed nN (rpm)		3000	3000	3000	3000	3000	3000
Rated current In(A)		0.88	1.2	1.55	2.93	3.9	5.3
Max torque Tm(N.m)		0.56	1.11	1.92	3.81	7.17	9.54
Max current Im (A)		3.3	4.4	5	9.4	12.4	16
Max speed (rpm)		6000	6000	6000	6000	6000	5000
Standstill torque Ts(N.m)		0.176	0.352	0.71	1.4	2.63	3.5
Standstill current Is(A)		0.98	1.32	1.7	3.2	4.3	5.83
Torque constant Kt (N.m/A)		0.2	0.3	0.5	0.51	0.7	0.66
Rotary inertia Jm (Kg·cm^2)		0.023	0.044	0.17	0.274	0.9	1.027
		0.025(with brake)	0.046(with brake)	0.174(with brake)	0.29(with brake)	0.95(with brake)	1.19(with brake)
Brake parameter	Brake holding torque T(N.m)	0.32	0.32	2	2	4	4
	Rated power（W）	6.1	6.1	7.6	7.6	11.5	11.5
	Rated voltage（VDC）	24	24	24	24	24	24
Insulation class		F	F	F	F	F	F
Max radial force Fr(N)		78	78	180	180	335	335
Max axial force Fa(N)		54	54	90	90	167.5	167.5
Weight G(Kg)		0.31	0.382	0.85	1.3	2	2.3
		0.5(with brake)	0.571(with brake)	1.2(with brake)	1.65（带抱闸）	2.6(with brake)	2.9(with brake)
Cooling method		Totally enclosed, self-cooling					
Protection level		IP65，IP54 at the shaft end (Note: add oil seal IP54 at the shaft end, no oil seal IP50)					
Operation environment	Temperature	－20～+40℃（no icing）					
	Humidity	20～80RH(no condensation)					
	Ambient environment	Keep away from corrosion, flammable gases, oil droplets, dust					
	Altitude	The rated working altitude is less than 1000 meters above sea level. When the working altitude is higher than 1000 meters, it is necessary to reduce the rated value by 1.5% for every 100 meters of elevation. The maximum working altitude is 2000 meters above sea level.					

Note: ■ = A:Without brake
B :With brake
▲ = L :RS485，Pulse
E :EtherCAT

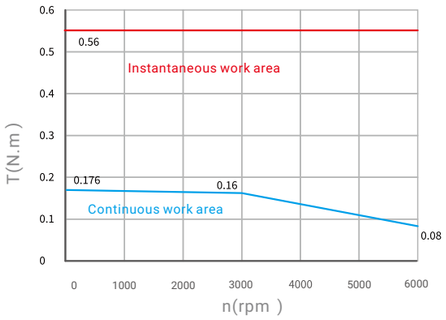
SMK servo motor technical parameter

Model parameter		Large inertia，130 flange				Large inertia，180 flange
Servo motor model		SMK130G-0085 -15Y■K-5LSR	SMK130G-0130 -15Y■K-5LSR	SMK130G-0180 -15Y■K-5LSR	SMK130G-0240 -15Y■K-5LSR	SMK180G-0300 -15Y■K-5HSR
Adapted drives		FD6P-▲A-L5R5 FD6P-▲A-H5R4	FD6P-▲A-H8R6	FD6P-▲A-H013	FD6P-▲A-H013	FD6P-▲A-H013
Rated voltage（VAC）		220	220	220	220	380
Rated power Pn(W)		850	1300	1800	2400	3000
Rated torque Tn(N.m)		5.39	8.27	11.45	15.2	19.1
Rated speed nN (rpm)		1500	1500	1500	1500	1500
Rated current In(A)		5.4	7.5	10	12.9	10.6
Max torque Tm(N.m)		16.2	24.81	28.625	38	57.3
Max current Im (A)		16.7	22.8	26	32.3	31.8
Max speed (rpm)		3500	3500	3500	3500	4500
Standstill torque Ts(N.m)		5.929	9.1	12.595	16.72	21
Standstill current Is(A)		5.94	8.25	11	14.19	11.7
Torque constant Kt (Nm/A)		1.082	1.197	1.26	1.293	1.98
Rotary inertia Jm (Kg·cm^2)		11.56	17.17	22.85	30.37	46
		12.86(with brake)	18.47(with brake)	24.15(with brake)	31.67(with brake)	52.43(with brake)
Brake parameter	Brake holding torque T(N.m)	20	20	20	20	55
	Rated power（W）	23	23	23	23	36W
	Rated voltage（VDC）	24	24	24	24	24
Insulation class		F	F	F	F	F
Max radial force Fr(N)		686	686	686	686	1470
Max axial force Fa(N)		196	196	196	196	490
Weight G(Kg)		6.03	7.35	8.7	10.6	14
		7.38(with brake)	8.6(with brake)	9.95(with brake)	11.85(with brake)	18.9(with brake)
Cooling method		Totally enclosed, self-cooling				
Protection level		IP65，IP54 at the shaft end (Note: add oil seal IP54 at the shaft end, no oil seal IP50)				
Operation environment	Temperature	－20～+40℃（no icing）				
	Humidity	20～80RH(no condensation)				
	Ambient environment	Keep away from corrosion, flammable gases, oil droplets, dust				
	Altitude	The rated working altitude is less than 1000 meters above sea level. When the working altitude is higher than 1000 meters, it is necessary to reduce the rated value by 1.5% for every 100 meters of elevation. The maximum working altitude is 2000 meters above sea level.				

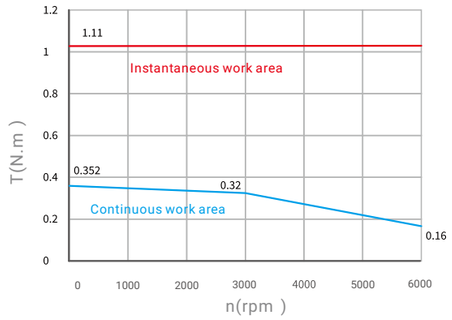
Note: ■ = A:Without brake
B :With brake
▲ = L :RS485，Pulse
E :EtherCAT

SMK series servo motor torque characteristic curve chart

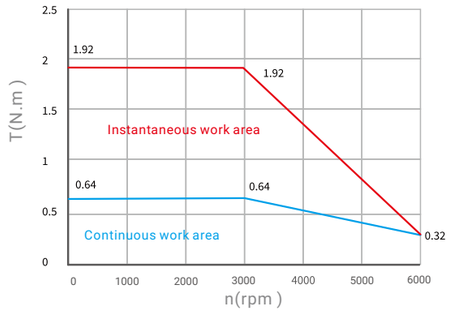
SMK40S-0005-30Y■K-5LSR 50W



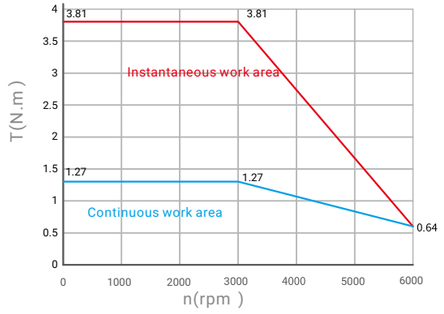
SMK40S-0010-30Y■K-5LSR 100W



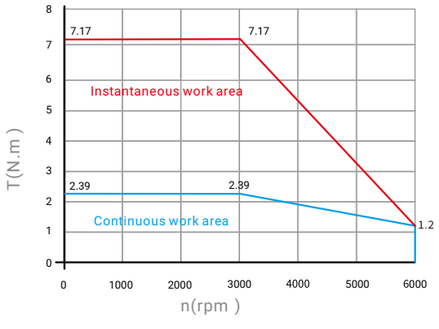
SMK60S-0020-30Y■K-5LSR 200W



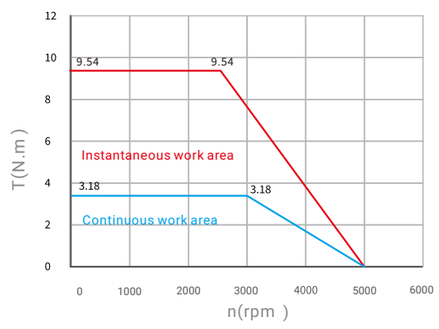
SMK60S-0040-30Y■K-5LSR 400W



SMK80S-0075-30Y■K-5LSR 750W

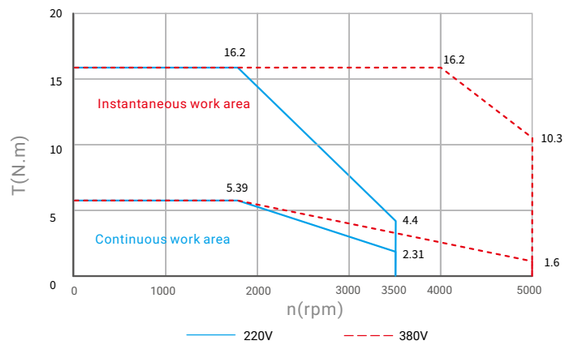


SMK80S-0100-30Y■K-5LSR 1000W

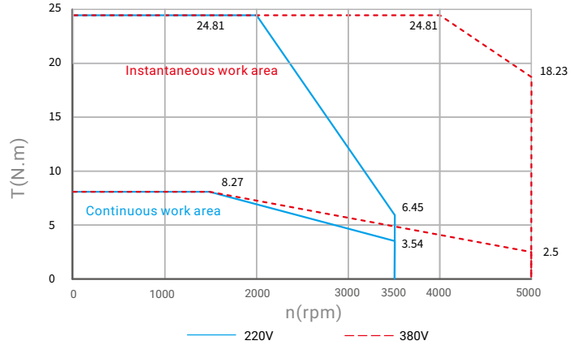


SMK series servo motor torque characteristic curve chart

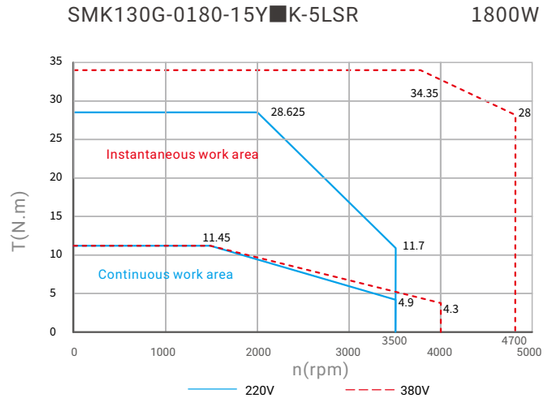
SMK130G-0085-15Y■K-5LSR 850W



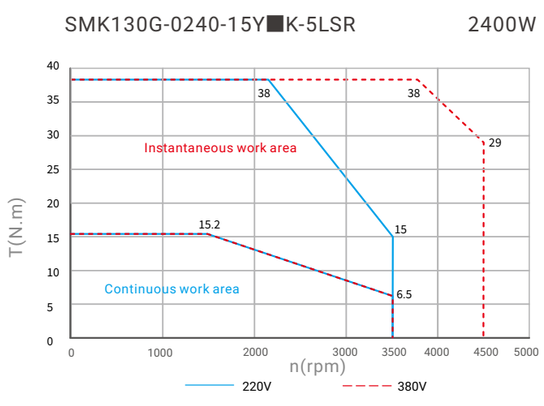
SMK130G-0130-15Y■K-5LSR 1300W



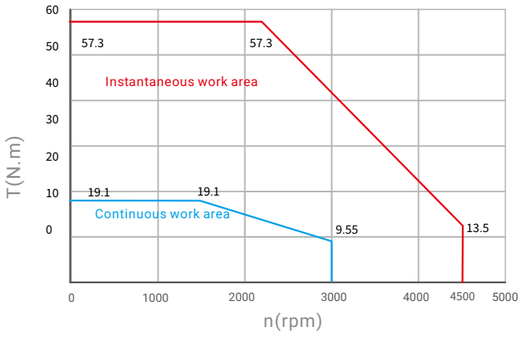
SMK130G-0180-15Y■K-5LSR 1800W



SMK130G-0240-15Y■K-5LSR 2400W

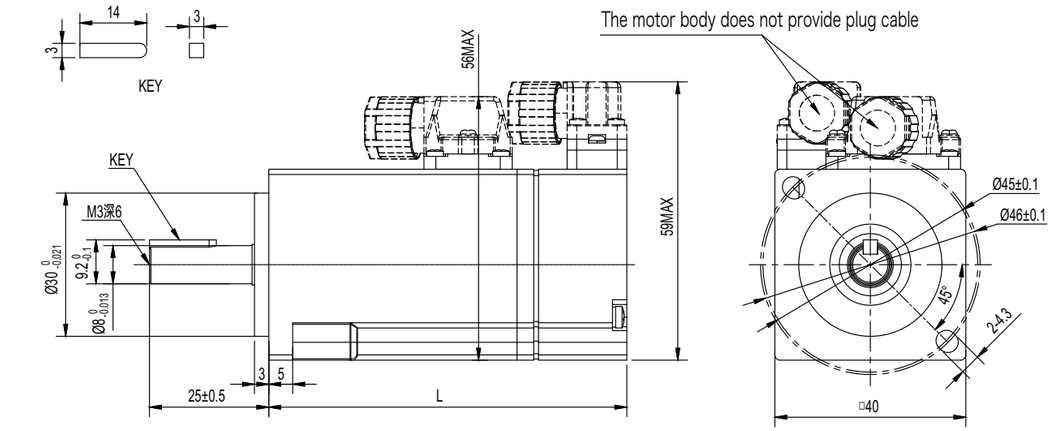


SMK180G-0300-15Y■K-5HSR 3000W



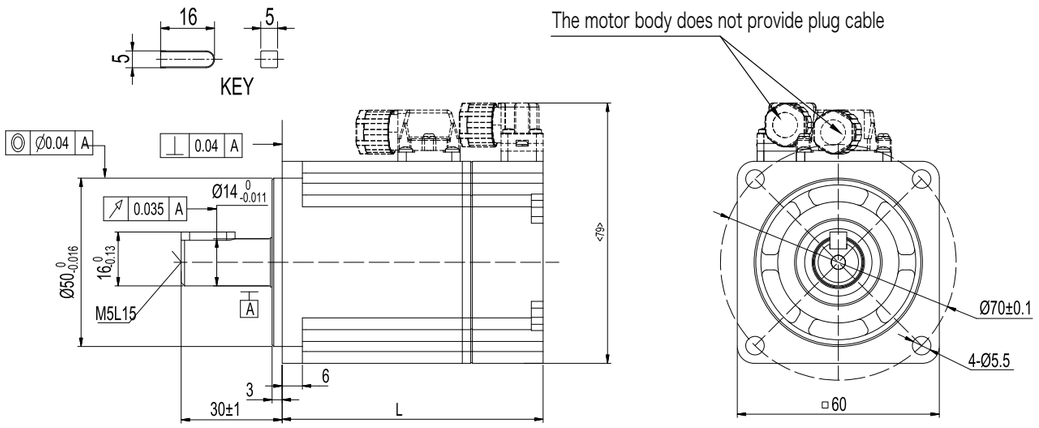
SMK servo motor dimension

SMK40S



SMK40S series model	With brake	Weight (kg)	Motor body size L (mm)
SMK40S-0005-30YAK-5LSR		0.31	62.5±1
SMK40S-0005-30YBK-5LSR	√	0.5	92.5±1
SMK40S-0010-30YAK-5LSR		0.382	75±1
SMK40S-0010-30YBK-5LSR	√	0.571	105±1

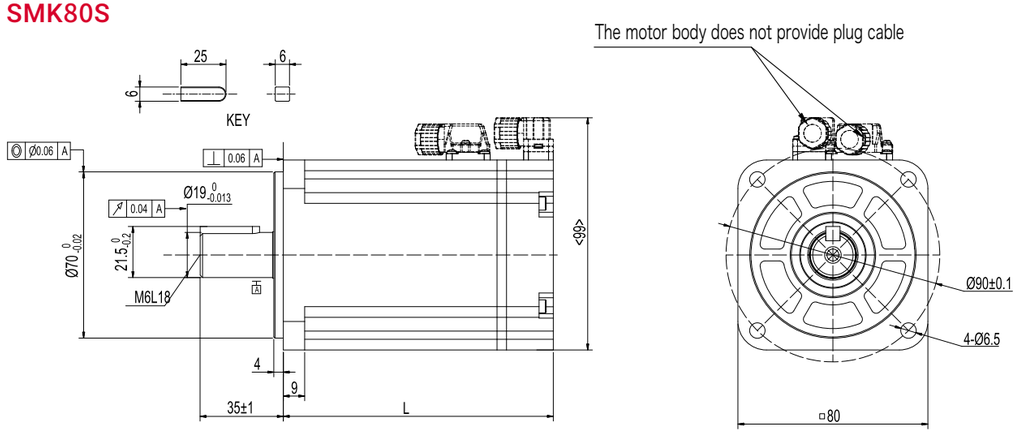
SMK60S



SMK60S series model	With brake	Weight (kg)	Motor body size L (mm)
SMK60S-0020-30YAK-5LSR		0.85	77±1.5
SMK60S-0020-30YBK-5LSR	√	1.2	109.1±1.5
SMK60S-0040-30YAK-5LSR		1.3	95±1.5
SMK60S-0040-30YBK-5LSR	√	1.65	127.1±1.5

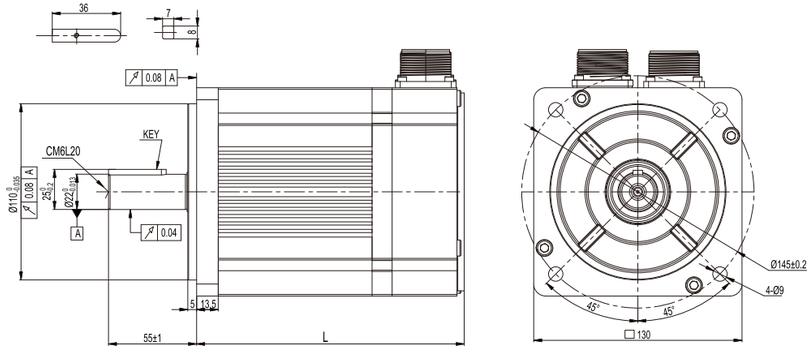
SMK servo motor dimension

SMK80S



SMK80S series model	With brake	Weight (kg)	Motor body size L (mm)
SMK80S-0075-30YAK-5LSR		2	103.7±1.5
SMK80S-0075-30YBK-5LSR	√	2.6	133.2±1.5
SMK80S-0100-30YAK-5LSR		2.3	113.7±1.5
SMK80S-0100-30YBK-5LSR	√	2.9	143.2±1.5

SMK130G

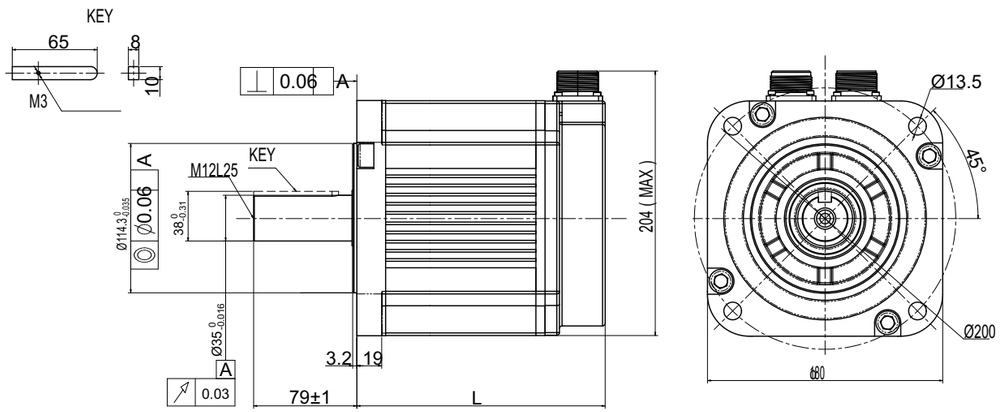


Note: The shaft diameter of SMK130G-0240-15Y■K-5LSR is 24 mm.

SMK130G series model	With brake	Weight (kg)	Motor body size L (mm)
SMK130G-0085-15YAK-5LSR		6.03	139±1.5
SMK130G-0085-15YBK-5LSR	√	7.38	155±1.5
SMK130G-0130-15YAK-5LSR		7.35	154±1.5
SMK130G-0130-15YBK-5LSR	√	8.6	170±1.5
SMK130G-0180-15YAK-5LSR		8.7	169±1.5
SMK130G-0180-15YBK-5LSR	√	9.95	185±1.5
SMK130G-0240-15YAK-5LSR		10.6	189±1.5
SMK130G-0240-15YBK-5LSR	√	11.85	205±1.5

SMK servo motor dimension

SMK180G

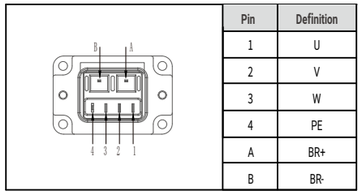


SMK180G series model	With brake	Weight (kg)	Motor body size L (mm)
SMK180G-0300-15YAK-5HSR		14	165±1
SMK180G-0300-15YBK-5HSR	√	18.9	215±1

Motor terminal connector pin definition

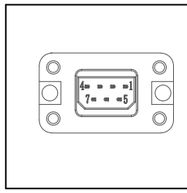
SMK series, 40, 60, 80 flange motor connectors

Power socket



Pin	Definition
1	U
2	V
3	W
4	PE
A	BR+
B	BR-

Encoder socket

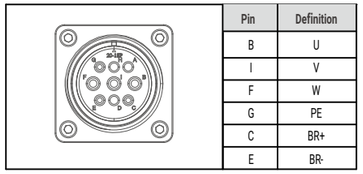


Pin	Multi-turn encoder definition
1	+5V
2	GND
3	VB+
4	VB-
5	SD+
6	SD-

Note: PINA and PINB pin signals are used for motor with brake

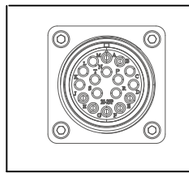
SMK series, 130 flange motor connectors

Power socket



Pin	Definition
B	U
I	V
F	W
G	PE
C	BR+
E	BR-

Encoder socket

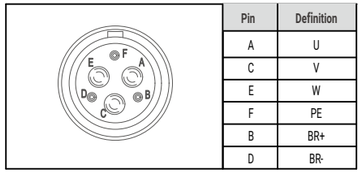


Pin	Multi-turn encoder definition
G	+5V
H	GND
E	VB+
F	VB-
A	SD+
B	SD-
J	PE

Note: PINC and PINE pin signals are used for motor with brake

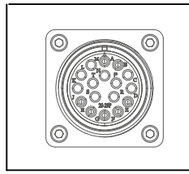
SMK series, 180 flange motor connectors

Power socket



Pin	Definition
A	U
C	V
E	W
F	PE
B	BR+
D	BR-

Encoder socket



Pin	Multi-turn encoder definition
G	+5V
H	GND
E	VB+
F	VB-
A	SD+
B	SD-
J	PE

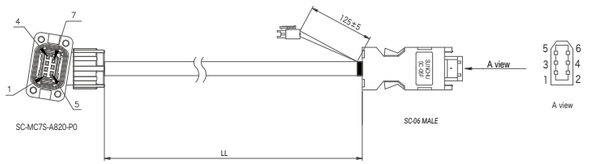
Note: PINC and PINE pin signals are used for motor with brake

Attachment

SMK series 40,60,80 flange motor encoder wire

■ E-D-QY-KA0-LL-F

Cable specification:
3P×26AWG with shielding, flexible cable (Specifications for cables of length 0-14m) .
1P×22AWG+2P×26AWG with shielding, flexible cable (Specifications for cables of length 15-39m)

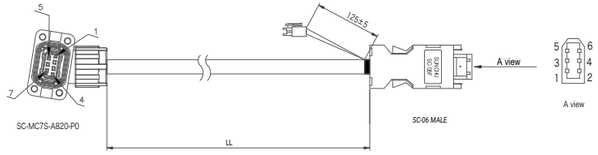


Signal	Motor end	Color	Drive end	
	SC-MC7S-A820-P0		Battery	SC-06 male
+5V	PIN1	Red	/	PIN1
GND	PIN2	Orange	/	PIN2
VB+	PIN3	Brown	PIN1	/
VB-	PIN4	Black	PIN2	/
SD+	PIN5	Blue	/	PIN5
SD-	PIN6	Purple	/	PIN6
Shield	PIN7	Shield	/	Metal button

Note:
Cable diameter:6.5±0.5mm
E-D-QY-KA0-LL-Faccessory package:E-KA and E-D

■ E-D-QY-KA1-LL-F

Cable specification:
3P×26AWG with shielding, flexible cable (Specifications for cables of length 0-14m).
1P×22AWG+2P×26AWG with shielding, flexible cable (Specifications for cables of length 15-39m)



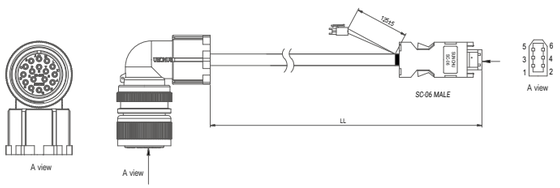
Signal	Motor end	Color	Drive end	
	SC-MC7S-A820-P0		Battery	SC-06 male
+5V	PIN1	Red	/	PIN1
GND	PIN2	Orange	/	PIN2
VB+	PIN3	Brown	PIN1	/
VB-	PIN4	Black	PIN2	/
SD+	PIN5	Blue	/	PIN5
SD-	PIN6	Purple	/	PIN6
Shield	PIN7	Shield	/	Metal button

Note:
Cable diameter:6.5±0.5mm
E-D-QY-KA1-LL-Faccessory package:E-KA and E-D

SMK series 130,180 flange motor encoder wire

■ E-D-QY-KR-LL-F

Cable specification:
3P×26AWG with shielding, flexible cable (Specifications for cables of length 0-14m).
1P×22AWG+2P×26AWG with shielding, flexible cable (Specifications for cables of length 15-39m)



Signal	Motor end	Color	Drive end	
	SUNCHU CMS3108A20-29SI aviation socket		Battery	SC-06 male
+5V	PIN G	Red	/	PIN1
GND	PIN H	Orange	/	PIN2
VB+	PIN E	Brown	PIN1	/
VB-	PIN F	Black	PIN2	/
SD+	PIN A	Blue	/	PIN5
SD-	PIN B	Purple	/	PIN6
Shield	PIN J	Shield	/	Metal button

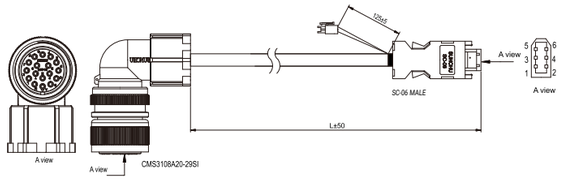
Note:
Cable diameter:6.5±0.5mm
E-D-QY-KR-LL-Faccessory package:E-D and E-KR

Attachment

SMK series 130,180 flange motor encoder wire

■ E-D-QY-KF-LL-F

Cable specification:
3P×26AWG with shielding, flexible cable (Specifications for cables of length 0-14m).
1P×22AWG+2P×26AWG with shielding, flexible cable (Specifications for cables of length 15-39m)



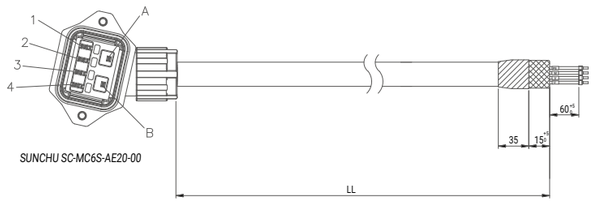
Signal	Motor end	Color	Drive end	
	SUNCHU CMS3108A20-29SI aviation socket		Battery	SC-06 male
+5V	PIN G	Red	/	PIN1
GND	PIN H	Orange	/	PIN2
VB+	PIN E	Brown	PIN1	/
VB-	PIN F	Black	PIN2	/
SD+	PIN A	Blue	/	PIN5
SD-	PIN B	Purple	/	PIN6
Shield	PIN J	Shield	/	Metal button

Note:
Cable diameter:6.5±0.5mm
E-D-QY-KR-LL-Faccessory package:E-D and E-KR

SMK series 40, 60, 80 flange motor power wire

■ M-A-6A-KA0-LL-FS

Cable specification:4C×20AWG with shielding, flexible cable

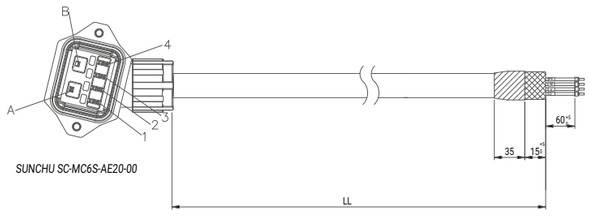


Signal	Motor end	Color
	SC-MC6S-AE20-00	
U	PIN1	White
V	PIN2	Red
W	PIN3	Black
PE	PIN4	yellow-green

Note:
Cable diameter:6.5±0.5mm
M-A-6A-KA0-LL-FSaccessory package:M-KA0

■ M-A-6A-KA1-LL-FS

Cable specification:4C×20AWG with shielding, flexible cable



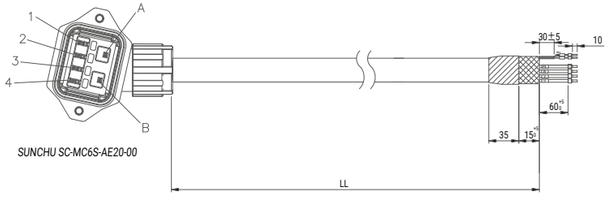
Signal	Motor end	Color
	SC-MC6S-AE20-00	
U	PIN1	White
V	PIN2	Red
W	PIN3	Black
PE	PIN4	yellow-green

Note:
Cable diameter:6.5±0.5mm
M-A-6A-KA1-LL-FSaccessory package:M-KA1

Attachment

M-A-6A-KA0-LL-BFS

Cable specification:4C×20AWG+1P*24AWG with shielding, flexible cable

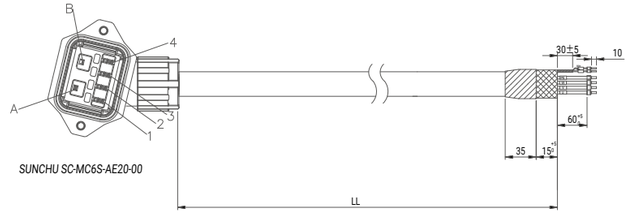


Signal	Motor end SC-MC6S-AE20-00	Color	Drive end C2505HM-02P
U	PIN1	White	\
V	PIN2	Red	\
W	PIN3	Black	\
PE	PIN4	yellow-green	\
BR+	PINA	Brown	PIN1
BR-	PINB	Blue	PIN2

Note:
Cable diameter:6.5±0.5mm
M-A-6A-KA0-LL-BFSaccessory package:M-KA0

M-A-6A-KA1-LL-BFS

Cable specification:4C×20AWG+1P*24AWG with shielding, flexible cable



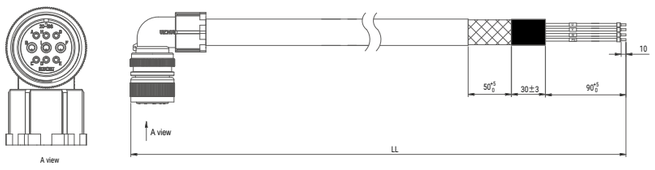
Signal	Motor end SC-MC6S-AE20-00	Color	Drive end C2505HM-02P
U	PIN1	White	\
V	PIN2	Red	\
W	PIN3	Black	\
PE	PIN4	yellow-green	\
BR+	PINA	Brown	PIN1
BR-	PINB	Blue	PIN2

Note:
Cable diameter:6.5±0.5mm
M-A-6A-KA1-LL-BFSaccessory package:M-KA1

SMK series 130 flange motor power wire

M-A-12A-KR0-LL-FS

Cable specification:4×16AWG with shielding, flexible cable



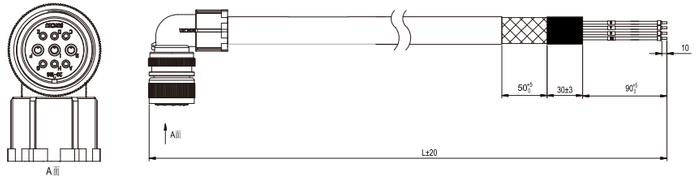
Signal	Motor end CMS3108A20-18SI aviation socket	Color
U	PINB	White
V	PINI	Red
W	PINF	Black
PE	PING	yellow-green

Note:
Cable diameter:8.5±0.5mm
M-A-12A-KR0-LL-FSaccessory package:M-KR0

Attachment

M-A-12A-KF0-LL-FS

Cable specification:4×16AWG with shielding, flexible cable

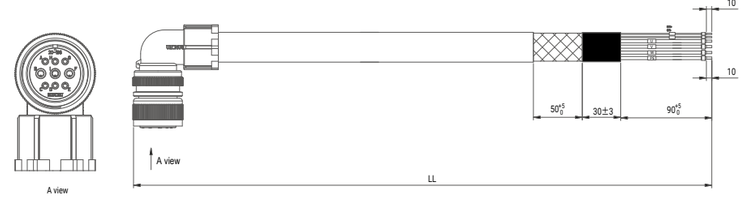


Signal	Motor end CMS3108A20-18SI aviation socket	Color
U	PINB	White
V	PINI	Red
W	PINF	Black
PE	PING	yellow-green

Note:
Cable diameter:8.5±0.5mm
M-A-12A-KR0-LL-FSaccessory package:M-KR0

M-A-12A-KR0-LL-BFS

Cable specification:4×16AWG+2×20AWG with shielding, flexible cable

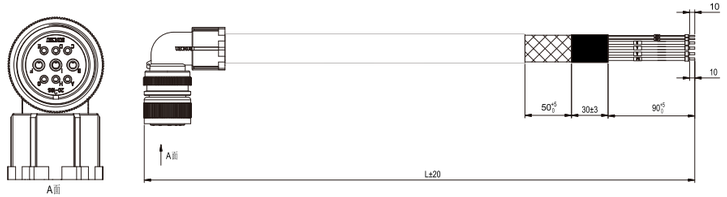


Signal	Motor end CMS3108A20-18SI aviation socket	Color
U	PINB	White
V	PINI	Red
W	PINF	Black
PE	PING	yellow-green
BR+	PINC	Brown
BR-	PINE	Blue

Note:
Cable diameter:10±0.5mm
M-A-12A-KR0-LL-BFSaccessory package:M-KR0

M-A-12A-KF0-LL-BFS

Cable specification:4×16AWG+2×20AWG with shielding, flexible cable



Signal	Motor end CMS3108A20-18SI aviation socket	Color
U	PINB	White
V	PINI	Red
W	PINF	Black
PE	PING	yellow-green
BR+	PINC	Brown
BR-	PINE	Blue

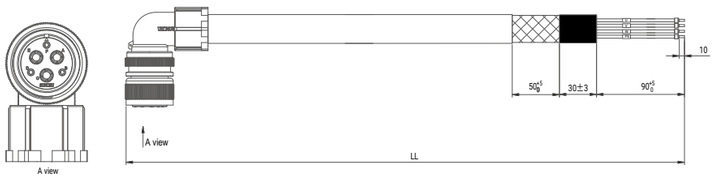
Note:
Cable diameter:10±0.5mm
M-A-12A-KR0-LL-BFSaccessory package:M-KR0

Attachment

SMK series 180 flange motor power wire

■ M-A-12A-KR1-LL-FS

Cable specification:4×16AWG with shielding, flexible cable

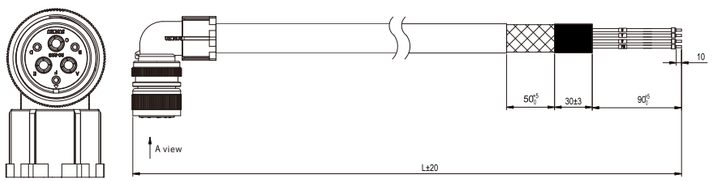


Signal	Motor end CMS3108A20-22SI aviation socket	Color
U	PINA	White
V	PINC	Red
W	PINE	Black
PE	PINF	yellow-green

Note:
Cable diameter:8.5±0.5mm
M-A-12A-KR1-LL-Faccessory package:M-KR1

■ M-A-12A-KF1-LL-FS

Cable specification:4×16AWG with shielding, flexible cable

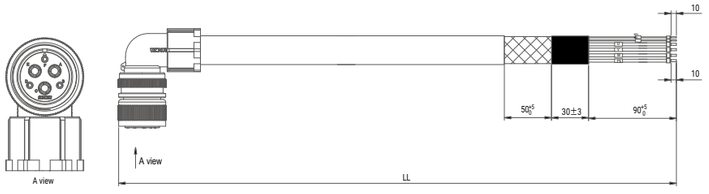


Signal	Motor end CMS3108A20-22SI aviation socket	Color
U	PINA	White
V	PINC	Red
W	PINE	Black
PE	PINF	yellow-green

Note:
Cable diameter:8.5±0.5mm
M-A-12A-KR1-LL-Faccessory package:M-KR1

■ M-A-12A-KR1-LL-BFS

Cable specification:4×16AWG+2×20AWG with shielding, flexible cable



Signal	Motor end CMS3108A20-22SI aviation socket	Color
U	PINA	White
V	PINC	Red
W	PINE	Black
PE	PINF	yellow-green
BR+	PINB	Brown
BR-	PIND	Blue

Note:
Cable diameter:10±0.5mm
M-A-12A-KR1-LL-BFSaccessory package:M-KR1

■ M-A-12A-KF1-LL-BFS

Cable specification:4×16AWG+2×20AWG with shielding, flexible cable



Signal	Motor end CMS3108A20-22SI aviation socket	Color
U	PINA	White
V	PINC	Red
W	PINE	Black
PE	PINF	yellow-green
BR+	PINB	Brown
BR-	PIND	Blue

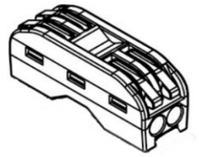
Note:
Cable diameter:10±0.5mm
M-A-12A-KR1-LL-BFSaccessory package:M-KR1

Attachment

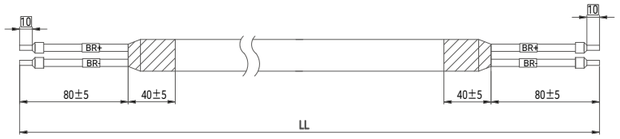
Brake extension cable

■ BRA-EXT-LL

Cable specification:2×20AWG without shielding



Quick-connect terminal



Signal	Color
BR+	Brown
BR-	Blue

Note:
Cable diameter:5.5±0.5mm

Battery

■ BAT-FD5

Note:BAT-FD5 battery is suitable for absolute encoder motor



Battery optional pack